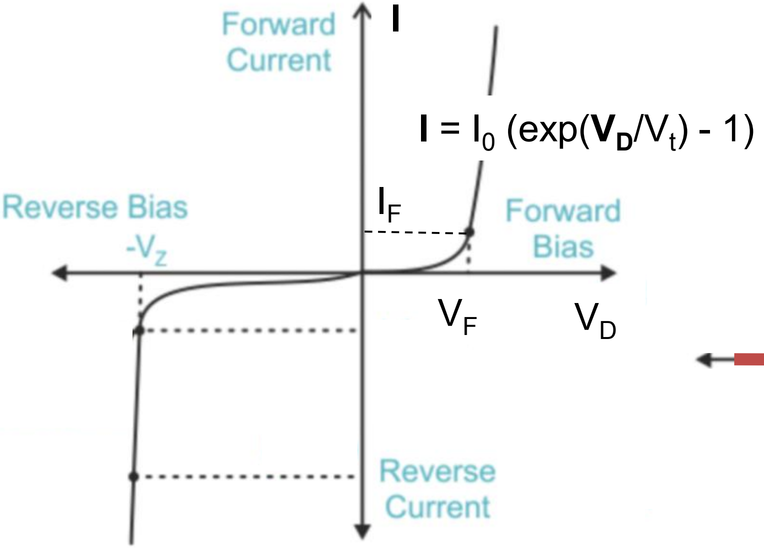
ES -116
Principles and Applications of Electrical Engineering

Semester II, 2025-26

Diode based Circuits for Different Utilities

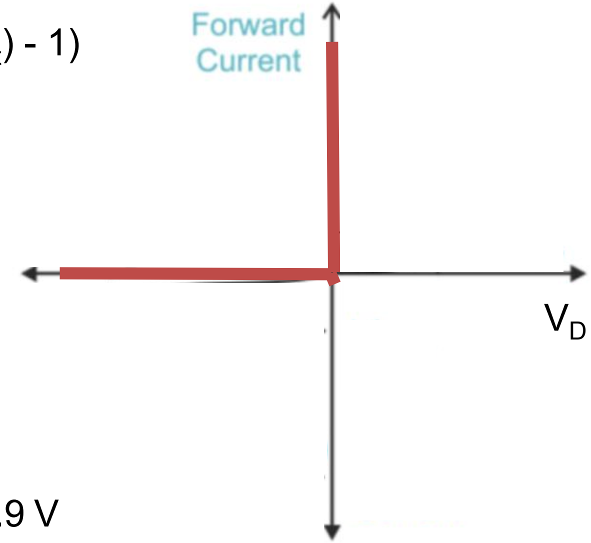
Recap: Diode Characteristics and different models



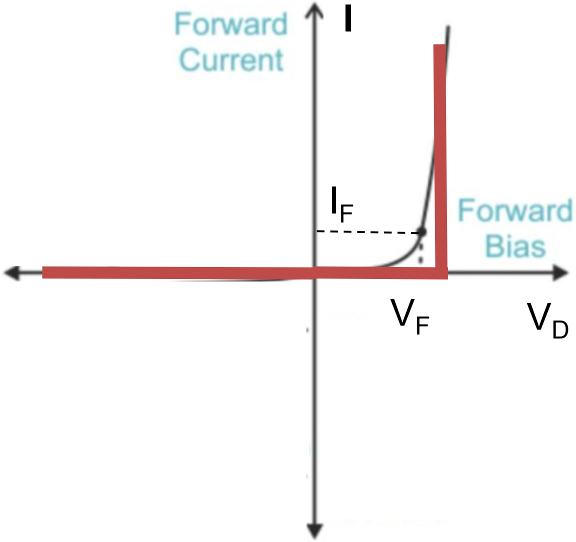
For typical silicon p-n diodes, V_F ~ 0.7-0.9 V and $-V_z$ ~ 50-200 V.

Specially designed Zener Diodes, V_F ~ 0.7-0.9 V and $-V_z$ ~ 3-8 V.

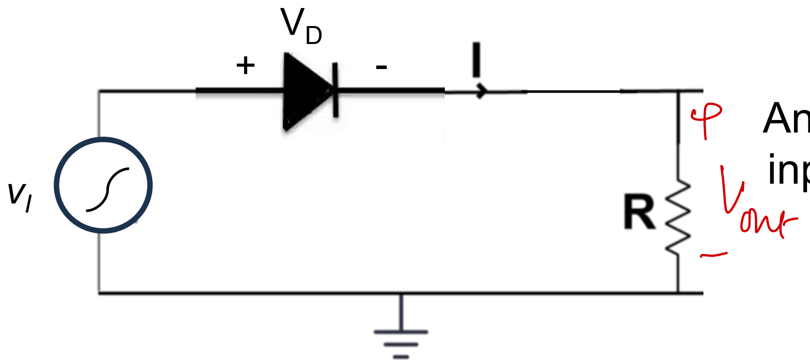
Ideal Model of the diode: Zero Drop across diode in FB



More practical Model of the diode: Constant voltage Drop across diode in FB

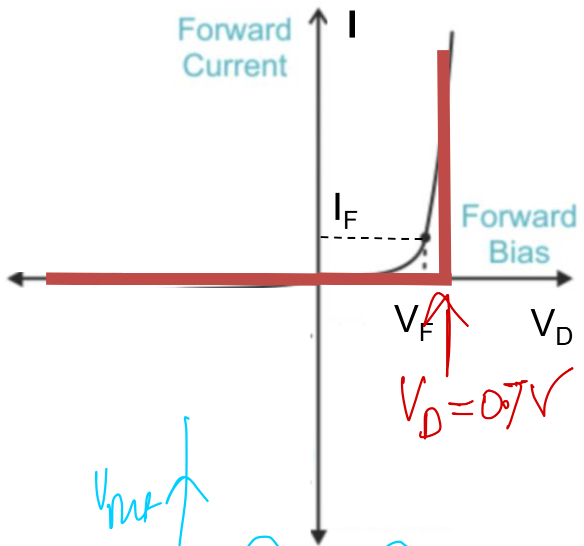


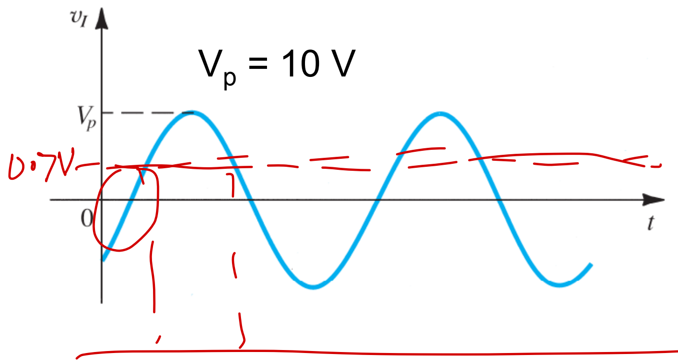
Recap: Using constant Voltage Drop Model

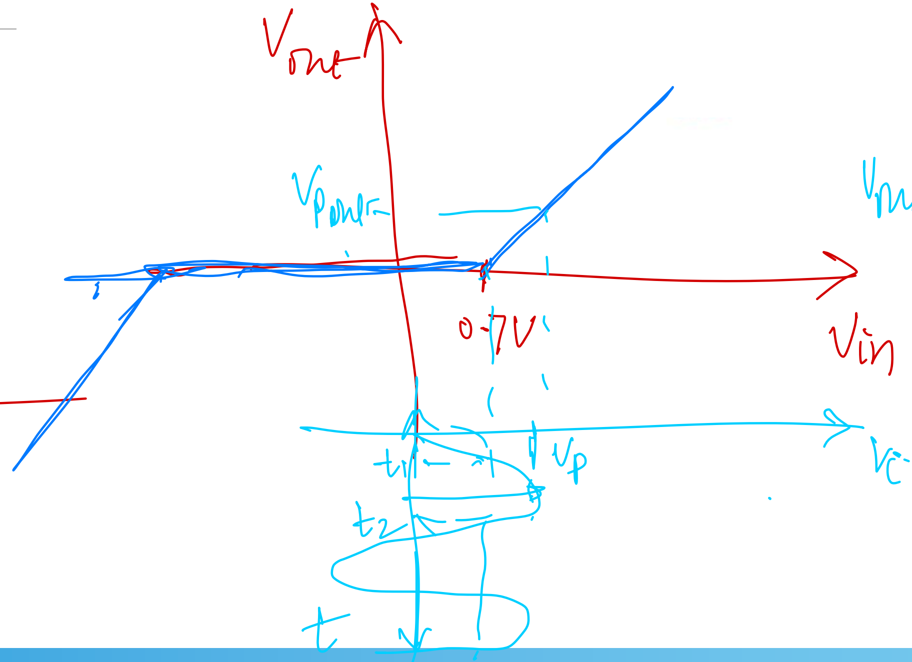


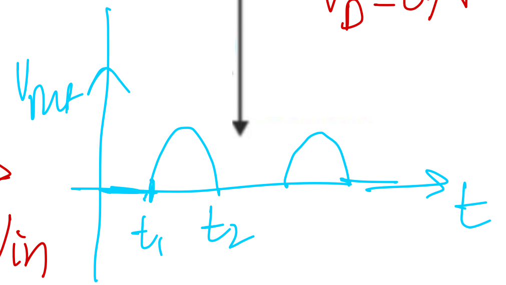
$$\mathbf{V_D} = V_{drop} \sim 0.7 \text{ V}$$

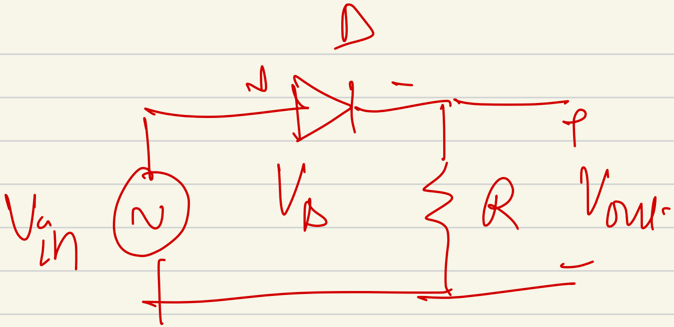
Analysing the diode circuit with AC input. Calculate the output voltage across the resistor.

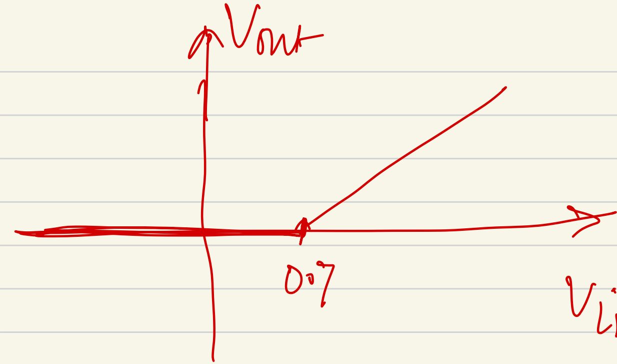




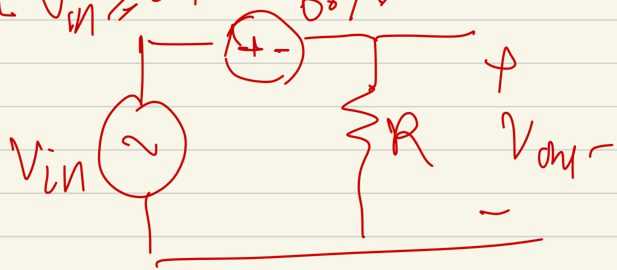




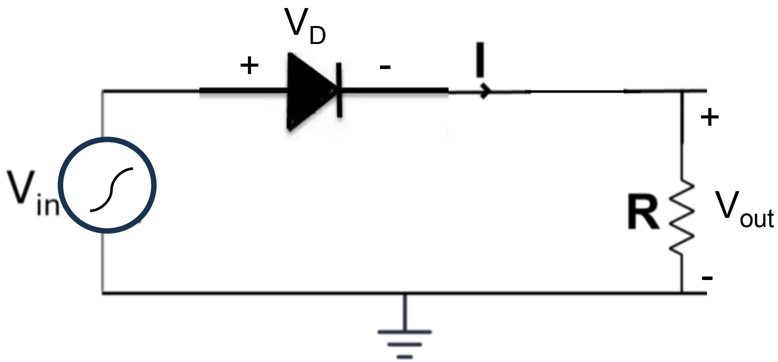


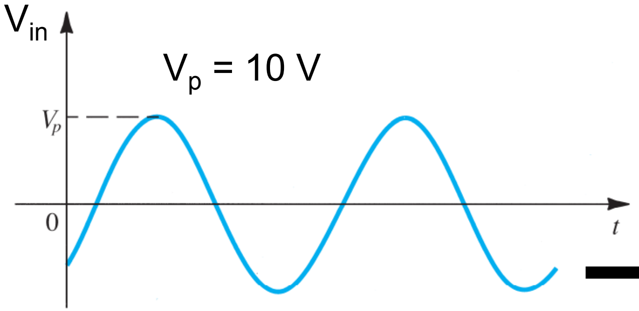


after $V_{in} \geq 0.7V$



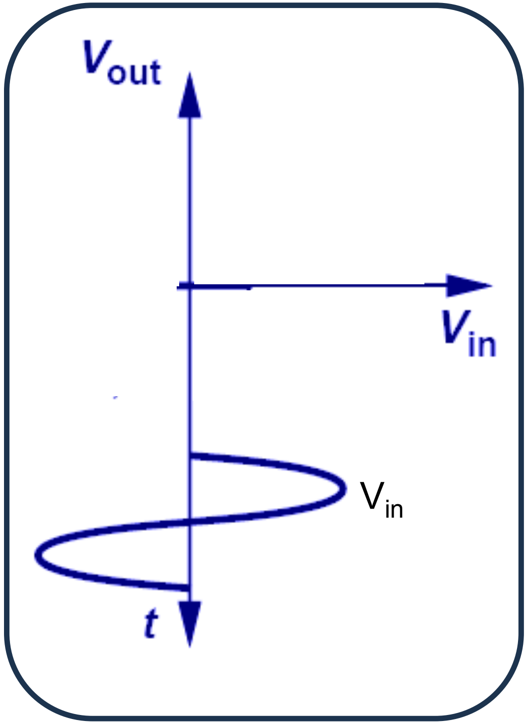
Drawing Circuit's Input-Output Characteristics

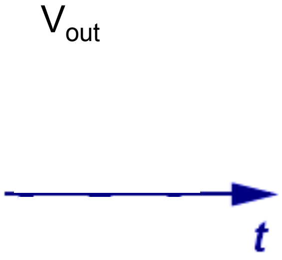




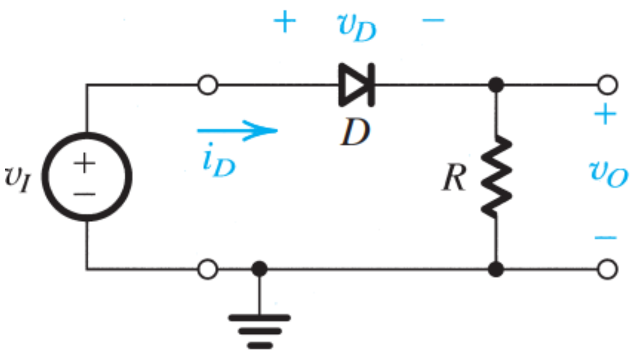
$\mathbf{V_D}$ = V_{drop} ~ 0.7 V

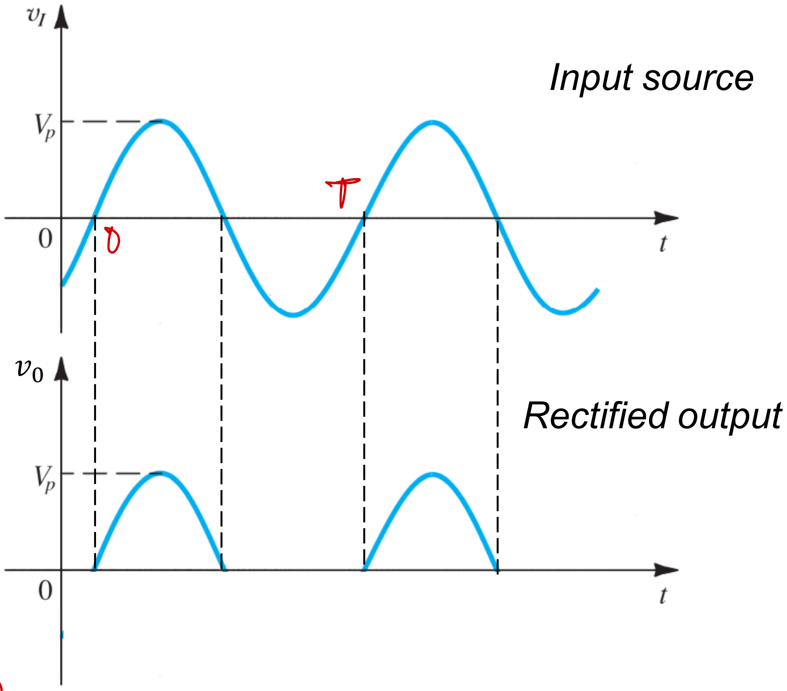
Analysing the diode circuit with AC input through drawing input-output characteristics (V_{out} vs V_{in}) of the circuit





A simple application: The rectifier





Calculate the average value of the rectified output?

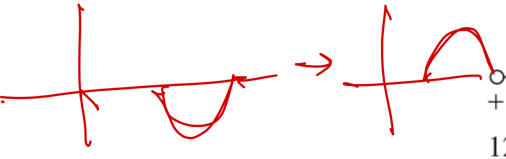
$$V_{avg} = \frac{1}{T}\int_0^T V_0(t)\,dt$$

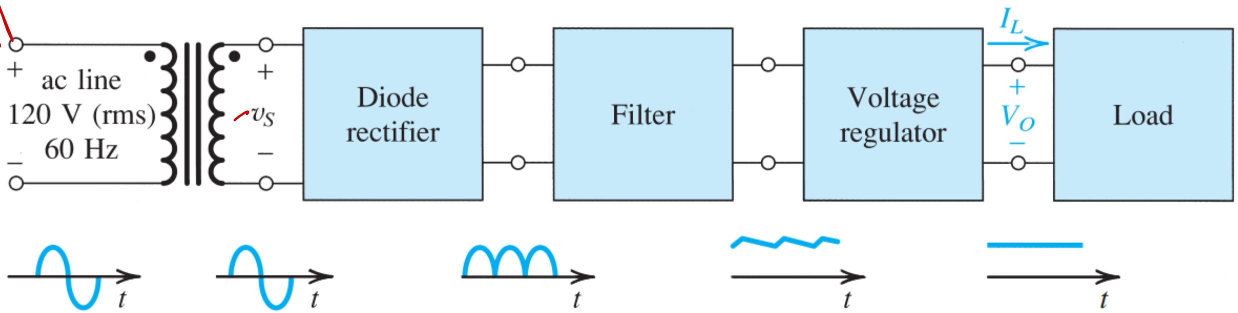
$$= \frac{1}{T}\int_0^{T/2} V_p \sin\omega t\,dt = \frac{V_p}{\pi}$$

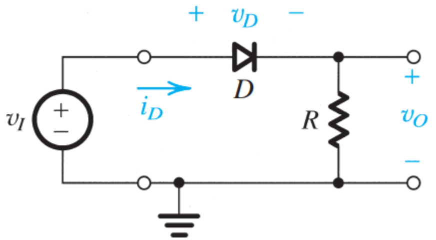
- **<u>Ideal diode model</u> (i.e. V_{drop} = 0)** can help us in quickly analyse circuits with diodes to understand its functionality.

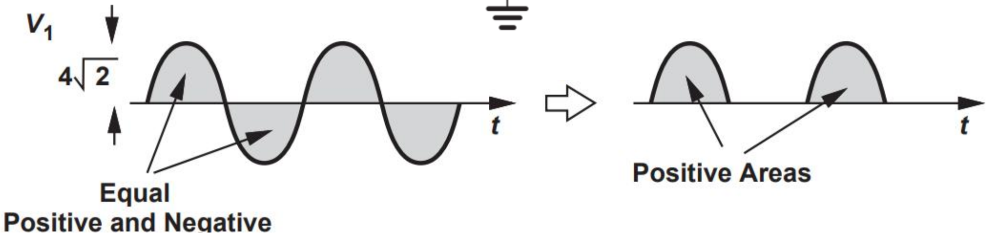
Is half-wave rectifier enough?

An application: AC to DC power supply

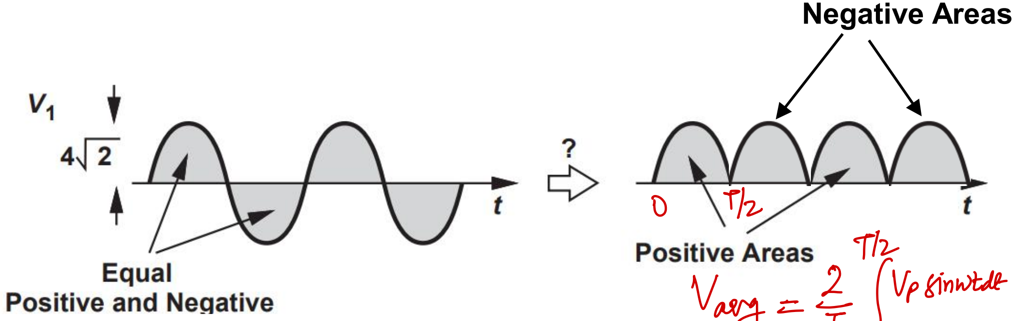








Maximizing the average voltage i.e. power in the rectified signal

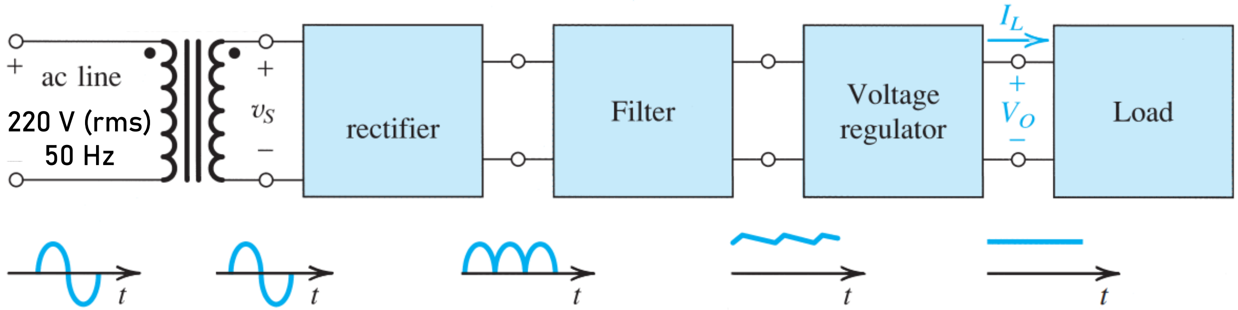


$$V_{avg} = \frac{2}{T}\int_0^{T/2} V_p \sin\omega t\,dt = \frac{2V_p}{\pi}$$

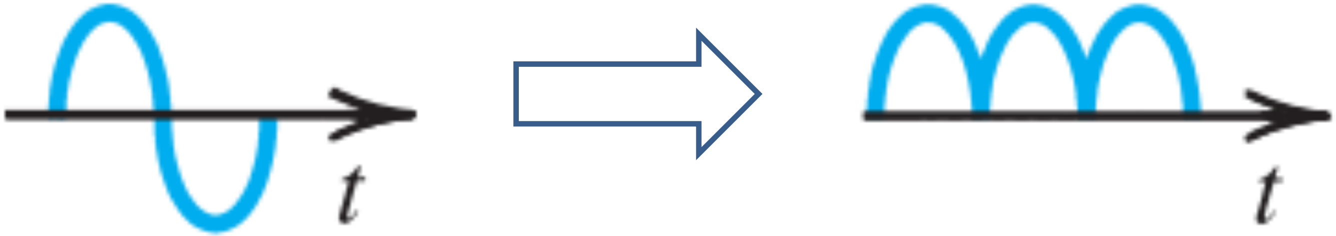
Practical Problem: Converting an AC to a DC

A cell phone adapter

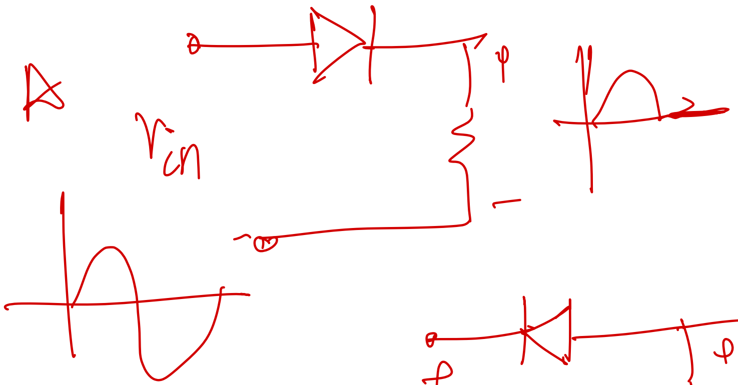
An application: AC to DC power supply

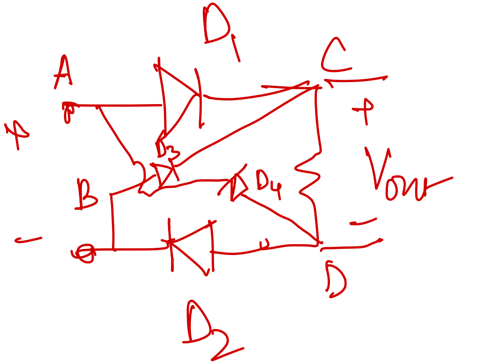


First step



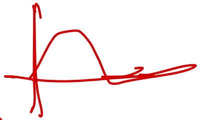
How to achieve this using the diode circuit?





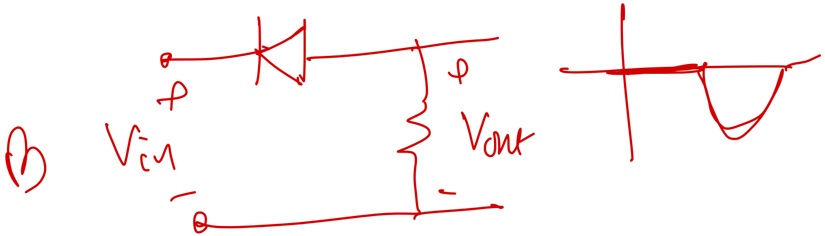
$V_{in} \to +ve$

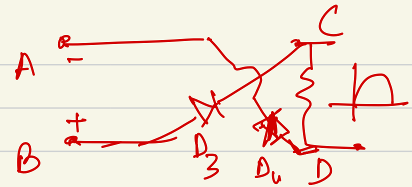
D_1 ON, D_2 ON

$V_{out} \to$ 

$V_{in} \to -ve$

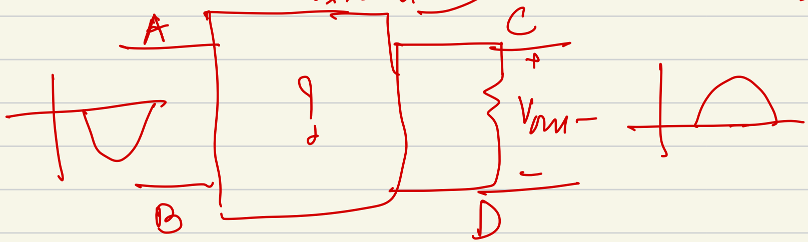
$D_1, D_2 \to$ OFF

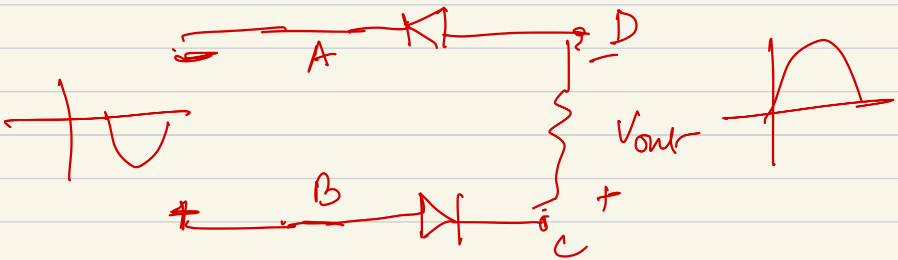


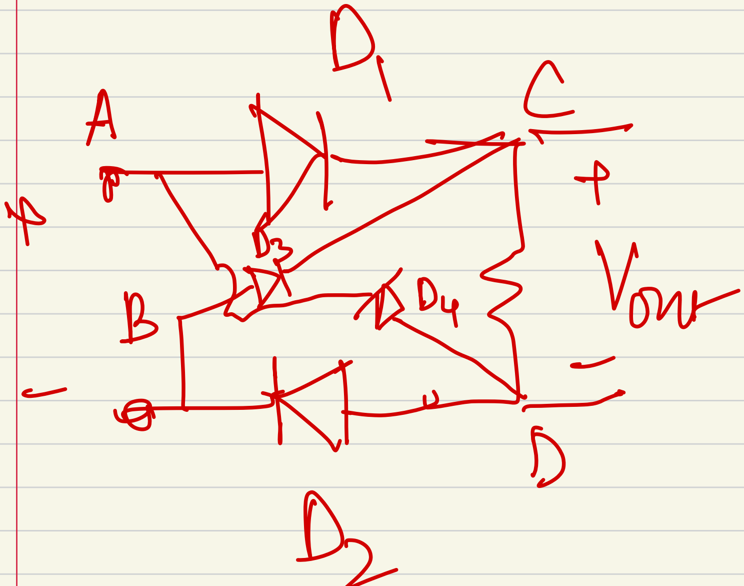


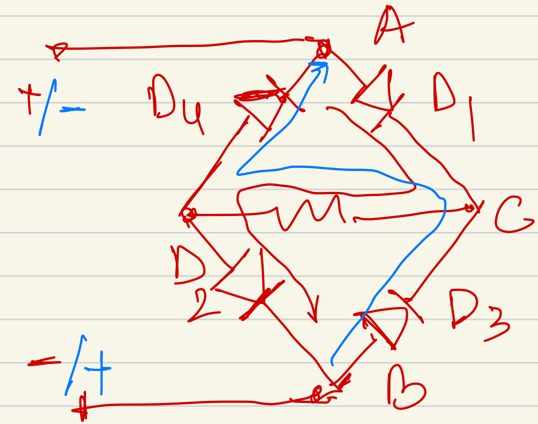
D_3 ON
D_4 ON

max. 2 diode



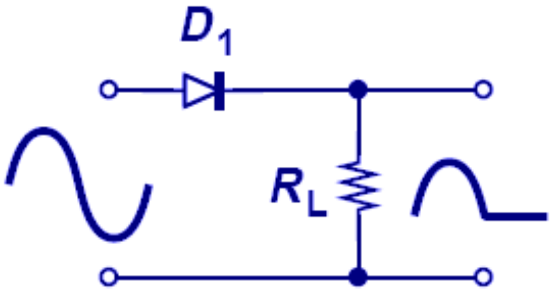


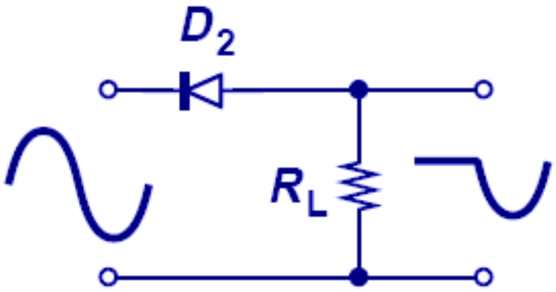


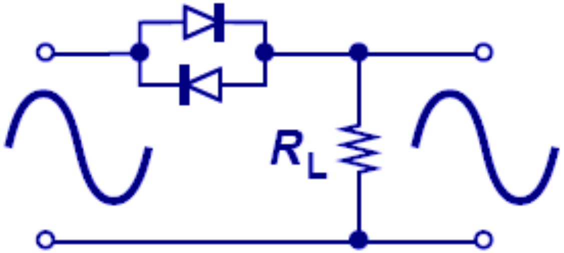


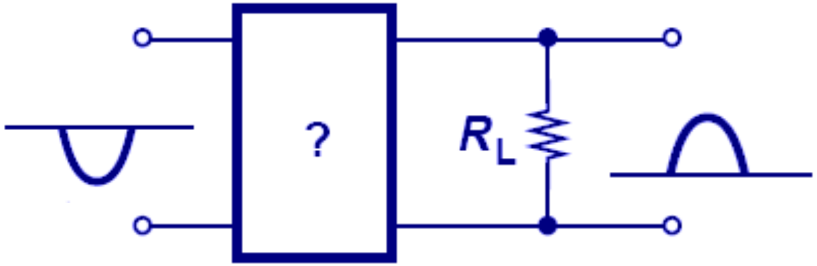
bridge rectifier

Evolution of full-wave rectifier

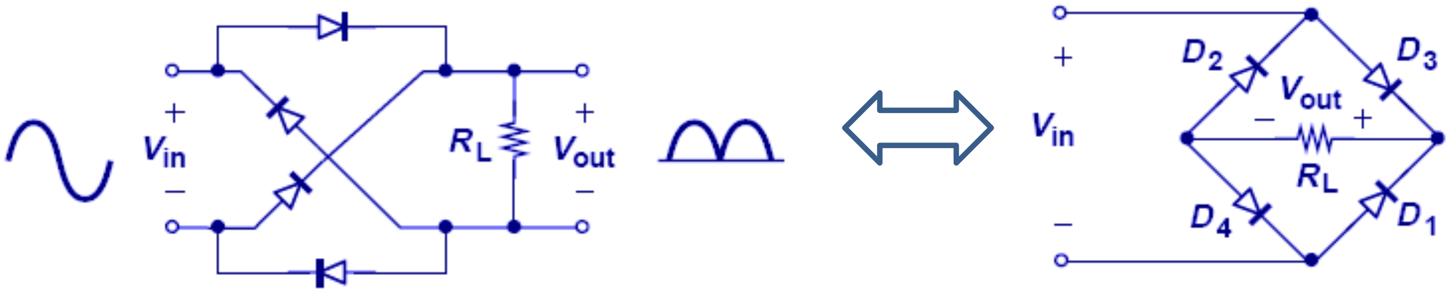






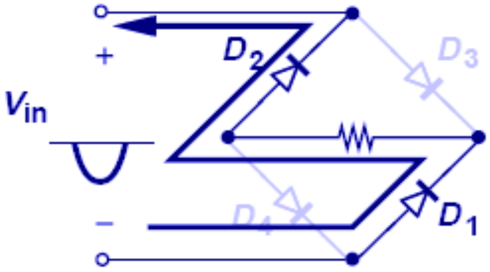


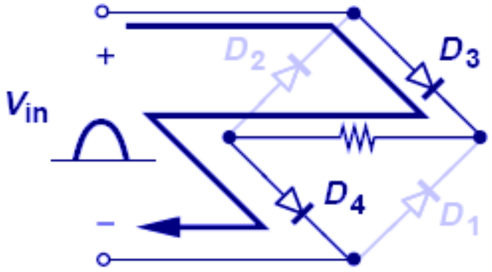
Full-wave rectifier: Bridge rectifier



- The figure above shows a full-wave rectifier, where D_1 and D_2 pass/invert the negative half cycle of input and D_3 and D_4 pass the positive half cycle.

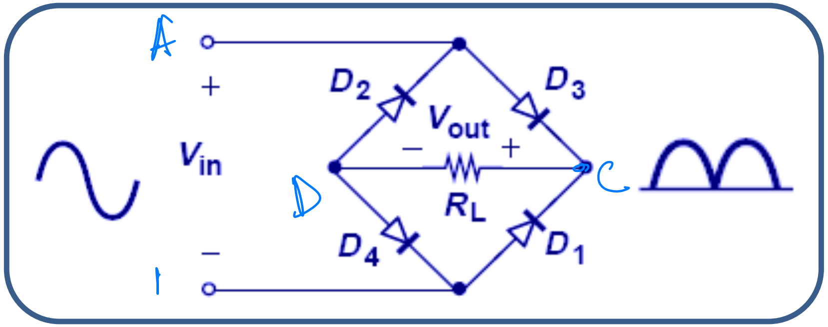
Full-wave rectifier: Bridge rectifier

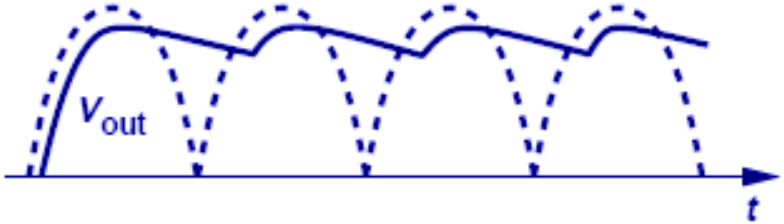




- The figure above shows a full-wave rectifier, where D_1 and D_2 pass/invert the negative half cycle of input and D_3 and D_4 pass the positive half cycle.

Next Goal





V_{in} V_p t_1 t_3 t_4

D_1 + V_C V_{out}

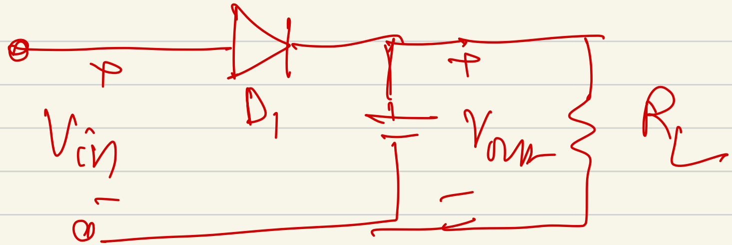
at $t = t_1$

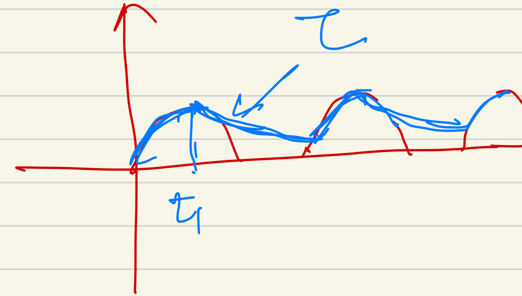
$V_C = V_p$

at $t = t_1 + 0^+$

$V_{D_1} = -ve$

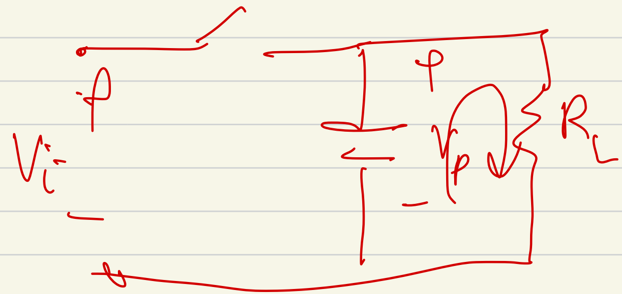
→ Off till





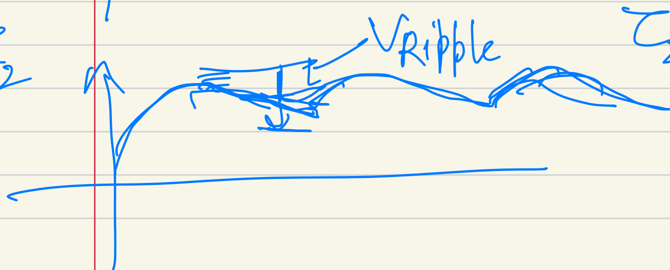
at $t = t_1 + 0_1$ D_1 is OFF

τ_1



V_{out}

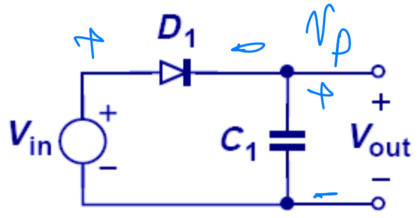
τ_2

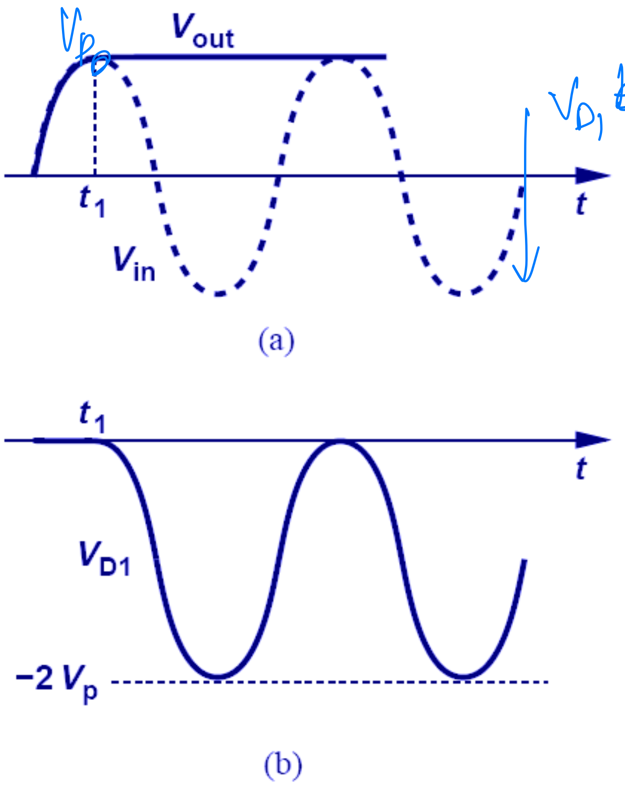


$\tau_2 >> \tau_1$

$\rightarrow$ ripples

Diode-capacitor circuit: Ideal model



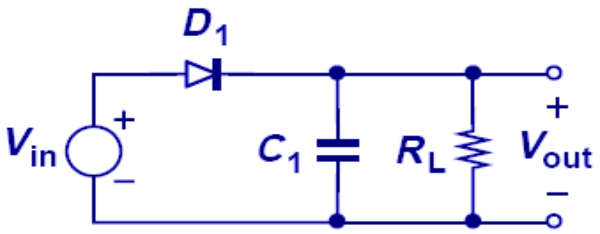


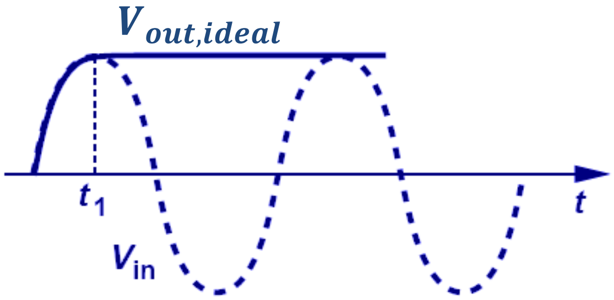
V_{D_1} by $-V_p$

at $t = t_1$, $V_{out} = V_p$

$V_{D_1} = V_{in} - V_p$

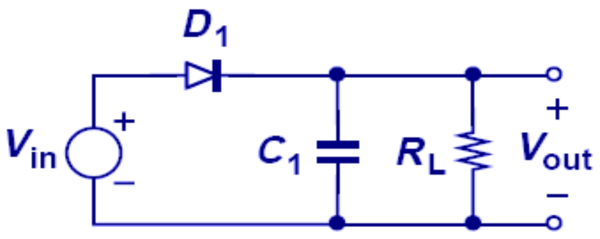
Diode-capacitor circuit: Practical model

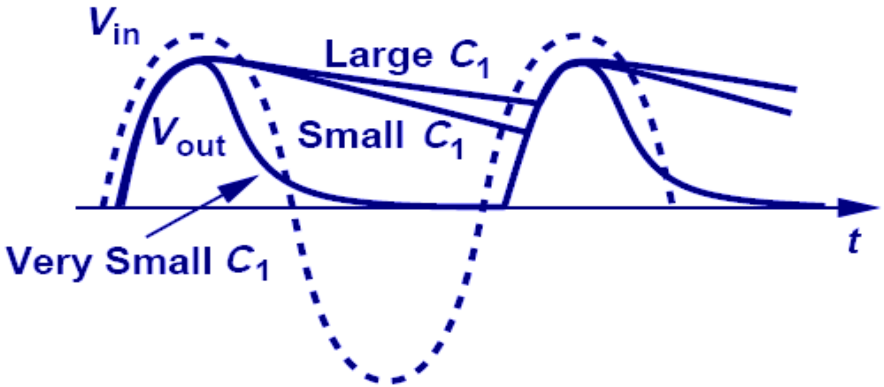




How would V_{out} look like for practical cases?

Behaviour for Different Capacitor Values





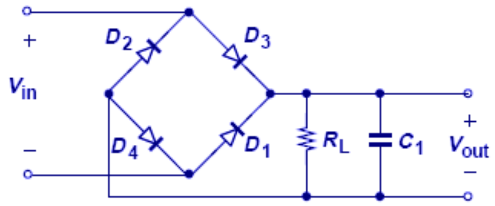
$$V_{out}(t) = (V_p - V_{D,on})\exp\frac{-t}{R_L C_1} \qquad 0 \le t \le T_{in}$$

$$V_{out}(t) \approx (V_p - V_{D,on})(1 - \frac{t}{R_L C_1}) \approx (V_p - V_{D,on}) - \frac{V_p - V_{D,on}}{R_L}\frac{t}{C_1}$$

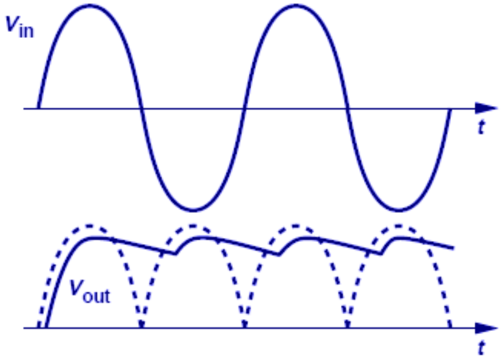
$$V_R \approx \frac{V_p - V_{D,on}}{R_L} \cdot \frac{T_{in}}{C_1} \approx \frac{V_p - V_{D,on}}{R_L C_1 f_{in}}$$

- For large C_1, V_{out} has small ripple.

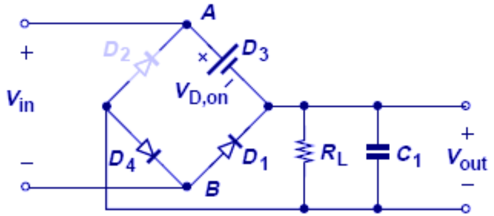
Complete full-wave rectifier



$$V_R \approx \frac{1}{2}\frac{V_P - V_{D,on}}{R_L C_1 f_{in}}$$

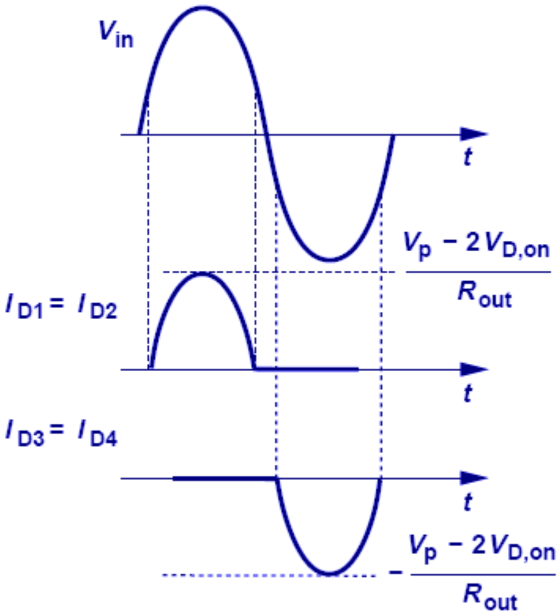


(a)



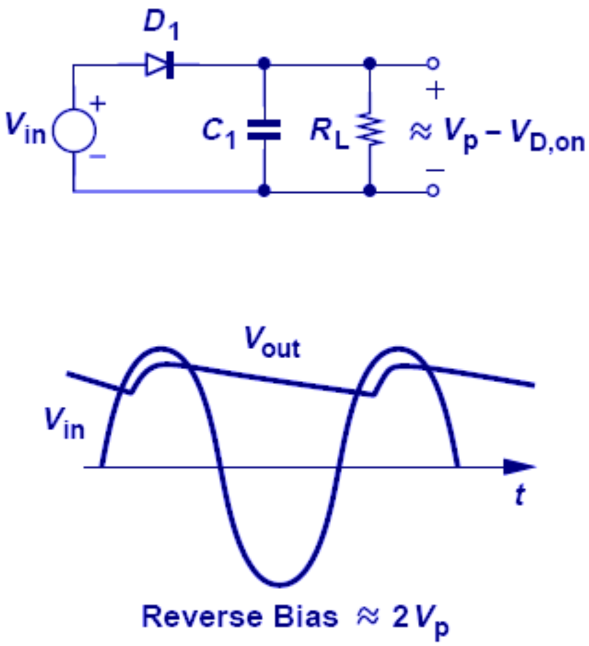
(b)

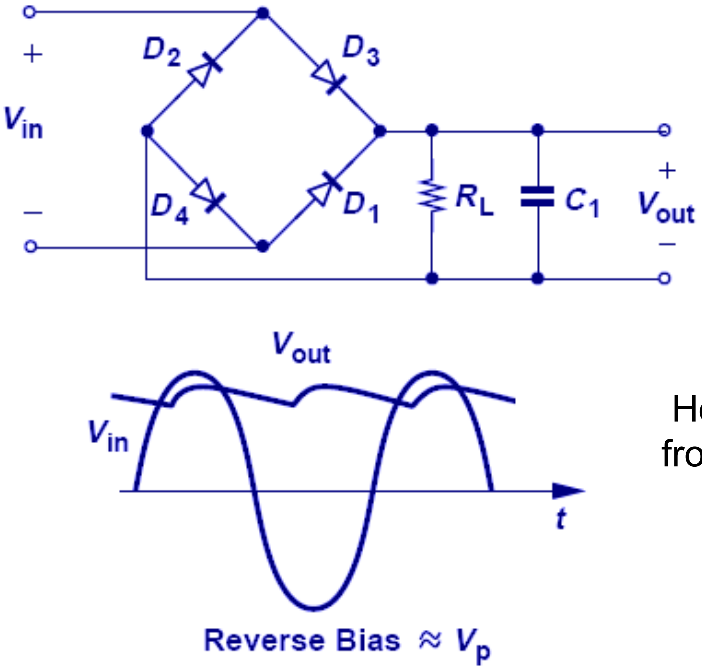
Current through each diode



- Since C_1 only gets half of the period to discharge, ripple voltage (V_R) is decreased by a factor of 2.
- Here, each diode is subjected to approximately one V_p reverse bias drop (versus $2V_p$ in half-wave rectifier).

Half-wave vs Full-wave rectifier



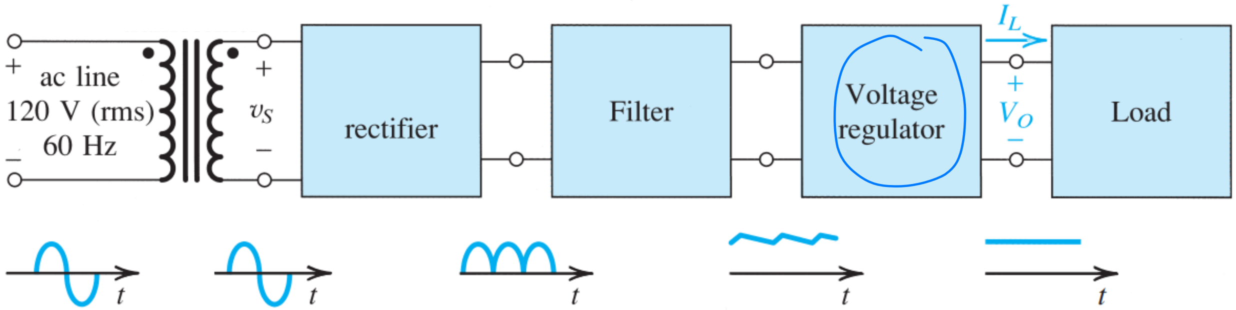


How to get rid of ripples from the rectified output?

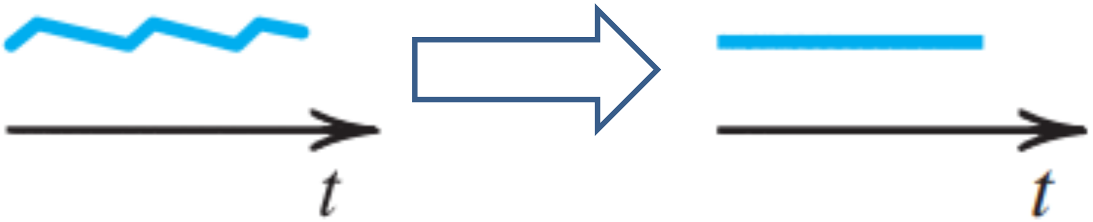
- Full-wave rectifiers are more suited to adapter and charger applications.

Problem: Converting an AC to a DC

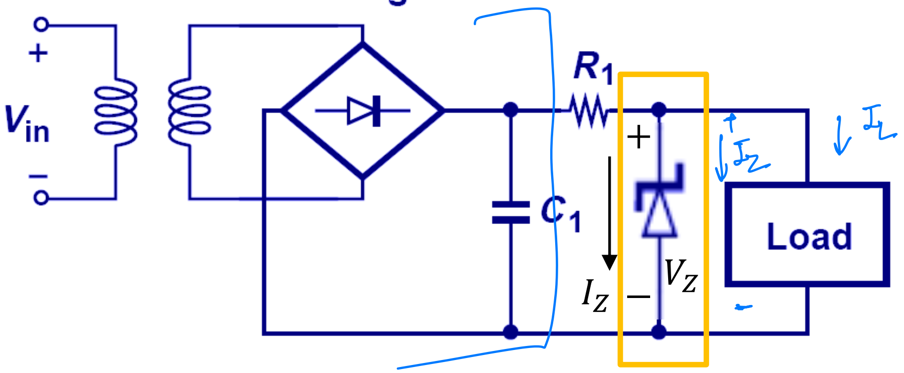
An application: AC to DC power supply



Last step



Voltage regulation with Zener diodes

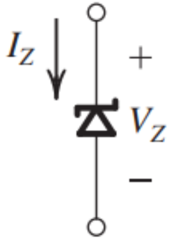


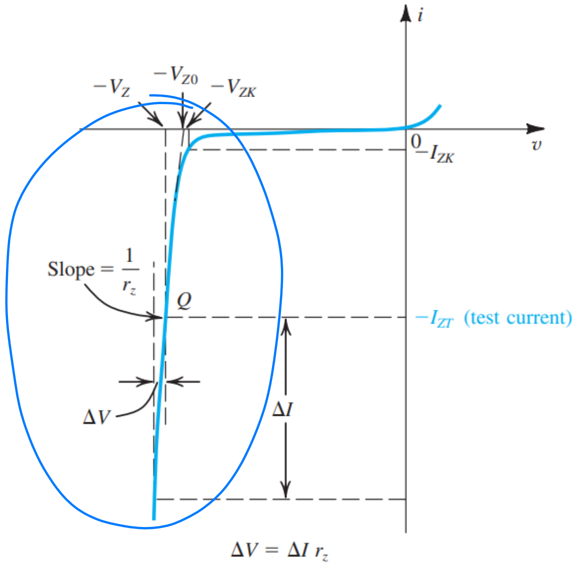
I_{ZK}: Knee current
V_{ZK}: Knee voltage

V_Z: Voltage across Zener diode at I_{ZT}
I_{ZT}: Test current specified in the data sheet

- Smaller r_d of Zener diodes in the breakdown region provides a regulated output voltage.

Circuit symbol of Zener diode





PRIMARY CHARACTERISTICS		
PARAMETER	**VALUE**	**UNIT**
V_Z range nom.	3.3 to 75	V
Test current I_{ZT}	3.3 to 76	mA
V_Z specification	Thermal equilibrium	
Circuit configuration	Single	

https://www.vishay.com/docs/85816/1n4728a.pdf

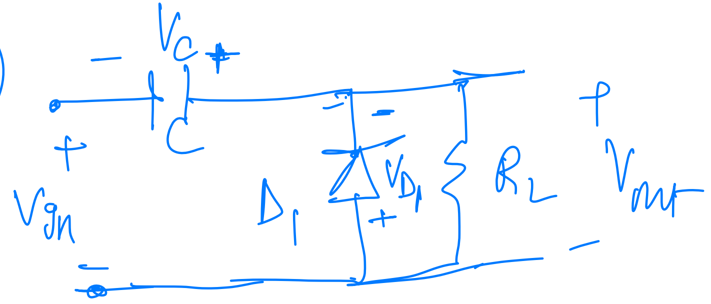
- It was introduced in the last problem of the last tutorial!

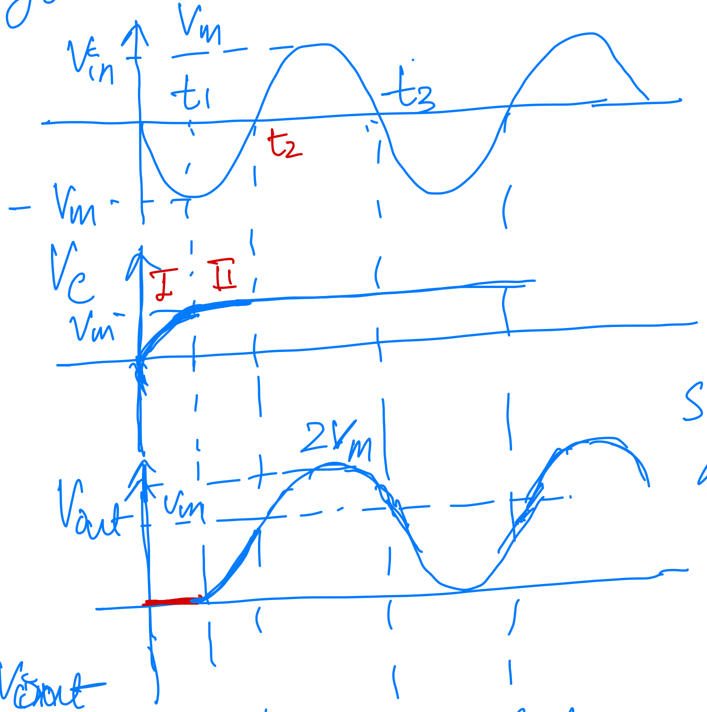
More Diode Based Circuits

- For example, practice clamper and other diode based circuits.

You have done this circuit in the lab!

(1)





Shifted up by V_m

assume $R_L C >> T_{in}$

$-ve$, $D_1 \rightarrow$ is ON till $t = t_1$

for $t_1 < t < t_2$, $V_{D_1} < 0$ as V_C is +ve $\Rightarrow$ D_1 is OFF

When D_1 is ON $\rightarrow$ $V_{out} = 0$ $(t < t_1)$

When D_1 is OFF $\rightarrow$ $V_{out} = V_{in} + V_C$

in region II, $V_{out} = V_{in} + V_C$ i.e V_{out} shifted by V_C (which is constant assuming very high time constant)

More Diode Based Circuits

- For example, practice clamper and other diode based circuits.

for $t > t_2$, when V_{in} is +ve $\rightarrow$ D_1 remains OFF as $V_{D_1} < 0$

$\Rightarrow V_{out} = V_{in} + V_C$, V_{out} shifted by V_C

for $t > t_3$ when V_{in} goes −ve, voltage across diode remains $= 0$, as $V_C = V_m$

even $V_{in} = -V_m$

as $V_{D_1} = -(V_{in} + V_C)$

D_1 remain off $\Leftarrow$

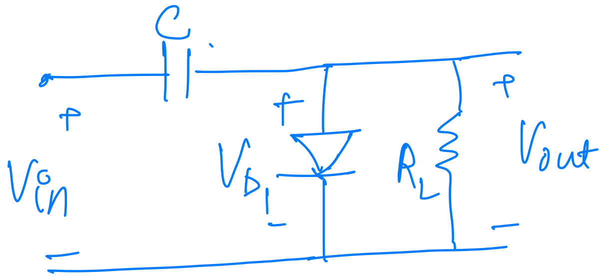
$V_{out} = V_{in} + V_C$

it continues.

More Diode Based Circuits

- For example, practice clamper and other diode based circuits.

Homework



Draw V_{D_1} & V_{out} for a sinusoidal V_{in}.